

§ 1.4 Review

1. Mathematical Induction

① Basic Step: $n=1$ holds

② Inductive Hypothesis: Suppose $n=k$ holds

③ WTS: $n=k+1$ holds

2. Lemma 1.25 (Pascal Triangle)

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k} = \binom{n}{k} + \binom{n}{k-1}$$

3. Binomial Theorem

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \quad (= \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k)$$

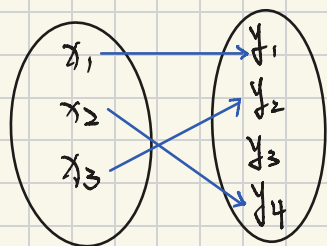
§ 1.5 Review

$f: X \rightarrow Y$

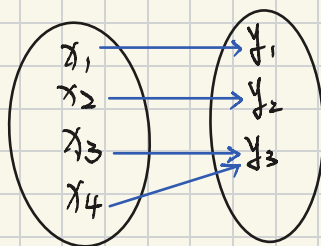
1. f is 1-1 $\Leftrightarrow x_1, x_2 \in X, f(x_1) = f(x_2) \Rightarrow \underline{x_1 = x_2}$

2. f is onto $\Leftrightarrow \forall \underline{y \in Y}, \exists \underline{x \in X}$ such that $y = f(x)$.

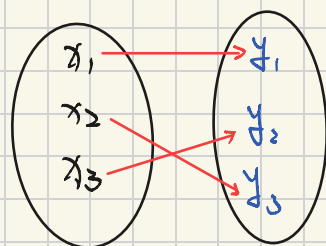
3. f is bijective $\Leftrightarrow f$ is 1-1 and onto.



1-1



onto



bijective

Image and Pre-Image

1. Image:

$$f(E) = \{y \in Y \mid y = f(x) \text{ for some } x \in E\}$$

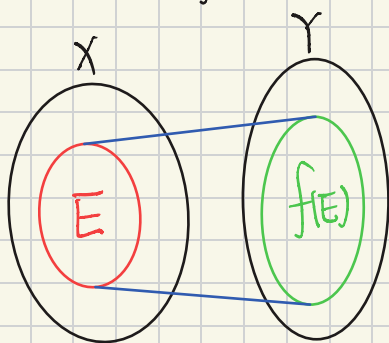
$$y = x^2$$

2. Pre-Image:

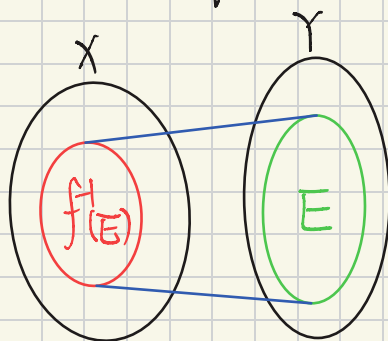
$$f^{-1}(E) = \{x \in X \mid f(x) = y \text{ for some } y \in E\}$$

$$E = [-1, 1]$$

$$f(E) = [0, 1]$$



Image



Pre-Image

3. Union and Intersection

$$\bigcup_{\alpha \in A} E_\alpha = E_1 \cup E_2 \cup E_3 \cup \dots$$

$$\bigcup_{\alpha \in A} E_\alpha = \{x \in X \mid x \in E_\alpha \text{ for some } \alpha \in A\}$$

$$\bigcap_{\alpha \in A} E_\alpha = \{x \in X \mid x \in E_\alpha \text{ for all } \alpha \in A\}$$

1.4.0. Decide which of the following statements are true and which are false. Prove the true ones and give counterexamples to the false ones.

- a) If $a \geq 0$ and $b \neq 0$, then $(a+b)^n \geq b^n$ for all $n \in \mathbf{N}$.
- b) If $a < 0 < b$, then $(a+b)^n \leq b^n$ for all $n \in \mathbf{N}$.
- c) If $n \in \mathbf{N}$ is even and both a and b are negative, then $(a+b)^n > a^n + na^{n-1}b$.
- d) If $a \neq 0$, then

$$\frac{1}{2^n} = \sum_{k=0}^n \binom{n}{k} \frac{(a-2)^{n-k}}{a^n 2^{n-k}}$$

for all $n \in \mathbf{N}$.

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

(c) If $n \in \mathbf{N}$ is even and $a, b < 0$, then $(a+b)^n > a^n + na^{n-1}b$.

pf By Binomial Thm, $(a+b)^n = a^n + na^{n-1}b + \sum_{k=2}^n \binom{n}{k} a^{n-k} b^k$

Since if n is even, then $n-k$ and k are both even or odd.

If both are even, $a^{n-k} \cdot b^k$ is $+$ * $+$ = $+$.

If both are odd, $a^{n-k} \cdot b^k$ is $-$ * $-$ = $+$.

Hence $a^{n-k} b^k$ is positive.

$$\text{Let } C = \sum_{k=2}^n \binom{n}{k} a^{n-k} b^k > 0$$

Therefore, $(a+b)^n = a^n + na^{n-1}b + C > a^n + na^{n-1}b \Rightarrow \text{True}$.

d) If $a \neq 0$, then

$$\frac{1}{2^n} = \sum_{k=0}^n \binom{n}{k} \frac{(a-2)^{n-k}}{a^n 2^{n-k}}$$

for all $n \in \mathbb{N}$.

pf Since $a \neq 0$, let $\frac{1}{2^n} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{a} + \frac{a-2}{2a}\right)^n$

Thus by Binomial Thm,

$$\begin{aligned} \frac{1}{2^n} &= \left(\frac{1}{a} + \frac{a-2}{2a}\right)^n = \sum_{k=0}^n \binom{n}{k} \frac{1}{a^k} \left(\frac{a-2}{2a}\right)^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} \frac{(a-2)^{n-k}}{a^k \cdot 2^{n-k} \cdot a^{n-k}} \\ &= \sum_{k=0}^n \binom{n}{k} \frac{(a-2)^{n-k}}{a^n 2^{n-k}} \Rightarrow \text{True.} // \end{aligned}$$

1.4.2. Use the Binomial Formula or the Principle of Induction to prove each of the following.

- a) $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$ for all $n \in \mathbf{N}$.
- b) $(a+b)^n \geq a^n + b^n$ for all $n \in \mathbf{N}$ and $a, b \geq 0$.
- c) $(1 + 1/n)^n \geq 2$ for all $n \in \mathbf{N}$.
- d) $\sum_{k=1}^n \binom{n}{k} = \sum_{k=0}^{n-1} 2^k$ for all $n \in \mathbf{N}$.

$$(a) \sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

$$\sum_{k=0}^n \binom{n}{k} (-1)^k \cdot 1^{n-k} = (-1+1)^n = 0^n = 0$$

$$(b) (a+b)^n \geq a^n + b^n, \forall n \in \mathbf{N} \text{ and } a, b \geq 0.$$

$$\begin{aligned} \text{pf } (a+b)^n &= a^n + \sum_{k=1}^{n-1} \binom{n}{k} a^k b^{n-k} + b^n \\ &= a^n + b^n + \sum_{k=1}^{n-1} \binom{n}{k} a^k b^{n-k} \end{aligned}$$

$$\text{Since } \forall k = 1, 2, \dots, n-1, a^k b^{n-k} \geq 0,$$

$$\text{let } C = \sum_{k=1}^{n-1} \binom{n}{k} a^k b^{n-k} \geq 0$$

$$\begin{aligned} \text{Hence } (a+b)^n &= a^n + b^n + \sum_{k=1}^{n-1} \binom{n}{k} a^k b^{n-k} \\ &= a^n + b^n + C \geq a^n + b^n \end{aligned}$$

$$\Rightarrow (a+b)^n \geq a^n + b^n, \forall n \in \mathbf{N}.$$

$\Rightarrow$ True.

$$(c) \left(1 + \frac{1}{n}\right)^n \geq 2, \forall n \in \mathbb{N}$$

Recall: $(a+b)^n \geq a^n + na^{n-1}b$, $\forall a, b \geq 0$ and $\forall n \in \mathbb{N}$.

pf. By $(a+b)^n \geq a^n + na^{n-1}b$, $\forall a, b \geq 0$ and $\forall n \in \mathbb{N}$.

$$\left(1 + \frac{1}{n}\right)^n \geq 1^n + n \cdot 1^{n-1} \cdot \frac{1}{n} = 1 + 1 = 2$$

$$(d) \sum_{k=1}^n \binom{n}{k} = \sum_{k=0}^{n-1} \binom{n}{k} \geq k, \forall n \in \mathbb{N}.$$

pf. By Binomial Thm, $\sum_{k=0}^n \binom{n}{k} = (1+1)^n - 1 = 2^n - 1$

$$\sum_{k=0}^{n-1} 2^k = 2^0 + 2^1 + 2^2 + \dots + 2^{n-1} = \frac{2^n - 1}{2 - 1} = 2^n - 1$$

Hence $\sum_{k=1}^n \binom{n}{k} = \sum_{k=0}^{n-1} 2^k$. //

法 2: MI

1. For $n=1$, $\sum_{k=1}^1 \binom{1}{k} = 1$ and $\sum_{k=0}^0 2^k = 2^0 = 1$, it holds.

2. Suppose for $n=m$, $\sum_{k=1}^m \binom{m}{k} = \sum_{k=0}^{m-1} 2^k$, we want to show that for $n=m+1$, $\sum_{k=1}^{m+1} \binom{m+1}{k} = \sum_{k=0}^m 2^k$.

By Lemma (Pascal), $\binom{m+1}{k} = \binom{m}{k-1} + \binom{m}{k}$, this implies

$$\begin{aligned} \sum_{k=1}^{m+1} \binom{m+1}{k} &= \sum_{k=1}^{m+1} \left[\binom{m}{k-1} + \binom{m}{k} \right] \\ &= \sum_{k=1}^{m+1} \binom{m}{k-1} + \sum_{k=1}^{m+1} \binom{m}{k} \\ &= \left(\sum_{k=2}^{m+1} \binom{m}{k-1} + \binom{m}{0} \right) + \left(\sum_{k=1}^m \binom{m}{k} + \underbrace{\binom{m}{m+1}}_0 \right) \end{aligned}$$

$$= \sum_{k=2}^{m+1} \binom{m}{k-1} + \sum_{k=1}^m \binom{m}{k} + \binom{m}{0}$$

$$= \sum_{\bar{c}=1}^m \binom{m}{\bar{c}} + \sum_{k=1}^m \binom{m}{k} + \binom{m}{0}$$

$\bar{c}=1, 2, \dots, m$

$\updownarrow$

(Let $\bar{c}=k-1$)

$$= \left(\binom{m}{0} + \sum_{\bar{c}=1}^m \binom{m}{\bar{c}} \right) + \sum_{k=1}^m \binom{m}{k}$$

$$= \sum_{\bar{c}=0}^m \binom{m}{\bar{c}} + \sum_{k=1}^m \binom{m}{k}$$

$$\sum_{k=1}^n \binom{n}{k} = (1+1)^n - 1 = 2^n - 1$$

$$= 2^m + \sum_{k=0}^{m-1} 2^k \leftarrow \text{(by inductive hypothesis: } \sum_{k=1}^m \binom{m}{k} = \sum_{k=0}^{m-1} 2^k \text{)}$$

$$= \sum_{k=0}^m 2^k$$

$$\sum_{k=1}^m \binom{m}{k} = \sum_{k=0}^{m-1} 2^k$$

Thus $\sum_{k=1}^{m+1} \binom{m+1}{k} = \sum_{k=0}^m 2^k \Rightarrow n=m+1$ holds.

Hence $\sum_{k=1}^n \binom{n}{k} = \sum_{k=0}^{n-1} 2^k$, proved by M.I. //

1.4.4 . Parts a) and c) of this exercise are used in Sections 2.4 and 5.1.

Prove that the following formulas hold for all $n \in \mathbb{N}$.

a) $\sum_{k=1}^n k = \frac{n(n+1)}{2}$

b) $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

c) $\sum_{k=1}^n \frac{a-1}{a^k} = 1 - \frac{1}{a^n}, a \neq 0$

d) $\sum_{k=1}^n (2k-1)^2 = \frac{n(4n^2-1)}{3}$

(c) $\sum_{k=1}^n \frac{a-1}{a^k} = 1 - \frac{1}{a^n}, a \neq 0.$

pf For $n=1$, $\sum_{k=1}^1 \frac{a-1}{a^k} = \frac{a-1}{a} = 1 - \frac{1}{a}$ holds.

Suppose for $n=M$, $\sum_{k=1}^M \frac{a-1}{a^k} = 1 - \frac{1}{a^M}$, we want to show $n=M+1$ holds.

If $n=M+1$, $\sum_{k=1}^{M+1} \frac{a-1}{a^k} = \sum_{k=1}^M \frac{a-1}{a^k} + \frac{a-1}{a^{M+1}}$

$$= 1 - \frac{1}{a^M} + \frac{a-1}{a^{M+1}}$$

$$= 1 + \left(\frac{-a + a - 1}{a^{M+1}} \right)$$

$$= 1 - \frac{1}{a^{M+1}}$$

Hence $\sum_{k=1}^n \frac{a-1}{a^k} = 1 - \frac{1}{a^n}, a \neq 0$ by M.I.

$$(d) \sum_{k=1}^n (2k-1)^2 = \frac{n(4n^2-1)}{3}$$

For $n=1$, $\sum_{k=1}^1 (2k-1)^2 = (2-1)^2 = 1$
 $\frac{1 \cdot (4-1)}{3} = 1$

$\Rightarrow$ it holds.

Suppose for $n=m$, $\sum_{k=1}^m (2k-1)^2 = \frac{m(4m^2-1)}{3}$, we want to show

$n=m+1$ holds.

$$2(m+1)-1 = 2m+2-1$$

$$\text{If } n=m+1, \sum_{k=1}^{m+1} (2k-1)^2 = \sum_{k=1}^m (2k-1)^2 + (2m+1)^2$$

$$4m^2-1 = (2m)^2-1^2$$

$$= (2m+1)(2m-1)$$

$$= \frac{m(4m^2-1)}{3} + (2m+1)^2$$

$$= \frac{m(2m+1)(2m-1) + 3(2m+1)^2}{3}$$

$$= \frac{(2m+1)(2m^2-m+6m+3)}{3}$$

$$= \frac{(2m+1)(2m^2+5m+3)}{3}$$

$$= \frac{(2m+1)(2m+3)(m+1)}{3}$$

$$= \frac{(m+1)(4m^2+8m+3)}{3}$$

$$= \frac{(m+1)(4m^2+8m+4-1)}{3}$$

$$= \frac{(m+1)(4(m+1)^2-1)}{3}$$

$$\begin{array}{r} \geq 3 \\ 1 \times 1 \end{array}$$

Hence $\sum_{k=1}^n (2k-1)^2 = \frac{n(4n^2-1)}{3}$ by M.I. //

1.4.5 . This exercise is used in Section 2.3. Prove that $0 \leq a < b$ implies $0 \leq \sqrt[n]{a} < \sqrt[n]{b}$ for all $n \in \mathbb{N}$.

~~P~~ ① For $n=1$, it is clear that $0 \leq a < b$

Suppose for $n=k$ holds. we want to show $n=k+1$ holds.

Since $0 \leq a^k < b^k$ and $0 \leq a < b$, by (7),

$$0 \leq a^k \cdot a < b^k \cdot b \Rightarrow 0 \leq a^{k+1} < b^{k+1}$$

Hence $0 \leq a^n < b^n$, $\forall n \in \mathbb{N}$ by M.I.

② Suppose if $\sqrt[n]{a} < \sqrt[n]{b}$ is false, then $\sqrt[n]{a} \geq \sqrt[n]{b} \geq 0$

$$\text{By ①, } 0^n \leq (\sqrt[n]{b})^n \leq (\sqrt[n]{a})^n$$

$$(\sqrt[n]{b})^n = (b^{\frac{1}{n}})^n = b$$

$$\Rightarrow b \leq a \quad \text{--- X ---}$$

Hence $0 \leq \sqrt[n]{a} < \sqrt[n]{b}$ if $0 \leq a < b$. //

$$0 \leq a < b \quad \text{and} \quad 0 \leq c < d \quad \text{imply} \quad ac < bd, \quad (7)$$

1.5.0. Decide which of the following statements are true and which are false. Prove the true ones and give counterexamples to the false ones.

a) Let $f(x) = \sin x$. Then the function

$$f : \left[\frac{\pi}{2}, \frac{3\pi}{2} \right] \rightarrow [-1, 1]$$

A.B.C.CX

is a bijection, and its inverse function is $\arcsin x$.

b) Suppose that A , B , and C are subsets of some set X and that $f : X \rightarrow X$. If $A \cap B \neq \emptyset$, then $f(A) \cap f(B \cup C) \neq \emptyset$.

c) Suppose that A and B are subsets of some set X . Then $(A \setminus B)^c = B \setminus A$.

d) If f takes $[-1, 1]$ onto $[-1, 1]$, then $f^{-1}(f(\{0\})) = \{0\}$.

pf (a) Since $\sin^{-1}x$ is $[-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$,

$$\text{thus } f^{-1}(x) = \pi + \sin^{-1}x.$$

$\Rightarrow$ false.

(b) If $A \cap B \neq \emptyset$, then $f(A) \cap f(B \cup C) = f(A) \cap f(B) \cup f(C)$

By thm 1.37, $(f(A) \cap f(B)) \cup f(C) \supseteq f(A \cap B) \cup f(C) \neq \emptyset$

Then $f(A \cap B) \cup f(C) \neq \emptyset$ by $f(A \cap B) \neq \emptyset$,

$$f(A) \cap f(B) \cup f(C) = f(A) \cap f(B \cup C) \neq \emptyset.$$

Hence it is true.